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Act as an expert CAT aptitude mentor and coach. I am preparing for the CAT examination, and I want you to decide the best learning path for me. Assess what I need to learn, determine the most effective order of topics, and create a structured study plan from basics to an advanced CAT level. Teach me interactively rather than just giving notes. Explain concepts clearly, build my intuition, and ensure I truly understand before moving on. Include shortcuts and faster solving techniques only after the fundamentals are clear. After every lesson: - Ask conceptual questions to check my understanding. - Give practice questions that gradually increase in difficulty. - Evaluate my answers and explain every mistake. - Identify patterns in my weaknesses and adjust future lessons accordingly. - Periodically revise older topics using spaced repetition. Whenever appropriate, include: - Time-saving strategies for CAT. - Multiple approaches to solving a problem. - Common traps and mistakes. - Tips for improving speed and accuracy under time pressure. Treat this as a long-term mentorship. Keep track of my progress, decide when I'm ready to move to the next topic, and continuously adapt the study plan based on my performance. Start by assessing my current aptitude level with a short diagnostic test, then create my personalized roadmap and begin teaching. First explain me each and everything about Aptitude concepts in detail with examples and don't skip any concept.

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Write a message...

+

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Claude is AI and can make mistakes. Please double-check responses.

CAT (Common Admission Test) tests three sections:

Section	Full Form	What It Tests
VARC	Verbal Ability & Reading Comprehension	RC passages, Para-jumbles, Para-summary, Odd sentence out
DILR	Data Interpretation & Logical Reasoning	Puzzles, Tables, Charts, Arrangements, Games & Tournaments
QA	Quantitative Ability	Arithmetic, Algebra, Geometry, Number Systems, Modern Math

CAT scoring reality: 99+ percentile in QA typically requires ~40-45/66 raw score. Speed + accuracy together — not just knowledge.

Diagnostic Test — Rules

- Attempt every question. If you don't know, write your best guess and mention you guessed.
- Show your working wherever possible — even rough thinking. This is where I spot your gaps.
- Don't use a calculator. CAT doesn't allow one.
- No time pressure right now — be honest, not fast.

SECTION 1 — Quantitative Ability (12 Questions)

Q1. Number Systems — Basics

Find the remainder when 2^{23} is divided by 7.

Q2. Arithmetic — Percentages

A shopkeeper marks his goods 40% above cost price and gives a discount of 20%. What is his profit percentage?

Q3. Arithmetic — Ratios & Mixtures

Two vessels contain milk and water in ratios 3:1 and 5:3 respectively. In what ratio should the mixtures be mixed so that the final ratio of milk to water is 2:1?

Q4. Arithmetic — Time, Speed & Distance

A train crosses a 200m platform in 20 seconds and crosses a man standing on the platform in 12 seconds. Find the speed of the train (in km/h).

Q5. Arithmetic — Time & Work

A can do a piece of work in 12 days. B is 50% more efficient than A. In how many days will they together finish the work?

Q6. Algebra — Linear & Quadratic

If $x + 1/x = 3$, find the value of $x^3 + 1/x^3$.

Q7. Algebra — Inequalities

For how many integer values of x is the expression $(x^2 - 5x + 6) / (x^2 - 4)$ negative?

Q8. Geometry — Circles & Triangles

In triangle ABC, angle B = 90° . If AB = 6 and BC = 8, find the radius of the circumscribed circle (circumcircle).

Q9. Geometry — Mensuration
A cone has base radius 7 cm and slant height 25 cm. Find the total surface area. (Use $\pi = 22/7$)

Q10. Number Theory — Factors & HCF/LCM
The HCF of two numbers is 12 and their LCM is 432. If one number is 48, find the other.

Q11. Modern Math — Permutation & Combination
In how many ways can the letters of the word **ARRANGE** be arranged so that the two R's are never together?

Q12. Modern Math — Probability
Two dice are thrown simultaneously. What is the probability that the sum of the numbers is a prime number?

SECTION 2 — DILR (3 Questions)

Q13. Logical Reasoning — Arrangement
Five people — A, B, C, D, E — sit in a row facing north.

- C sits at one of the extreme ends.
- B sits to the immediate left of D.
- A is not adjacent to C.
- E does not sit at an extreme end.

Who sits at the exact middle of the row?

Q14. Data Interpretation — Calculation Speed
A company's revenue (in ₹ crore) over 5 years:

Year	Revenue
2019	120
2020	132
2021	118
2022	145
2023	160

- (a) What is the percentage growth from 2019 to 2023?
(b) In which year was the percentage growth over the previous year the highest?

Q15. Logical Reasoning — Deduction
All managers are leaders.
Some leaders are visionaries.
No visionary is a lazy person.

Which of the following conclusions is **definitely** true?
(A) Some managers are visionaries.
(B) No manager is a lazy person.
(C) Some leaders are not lazy.
(D) All visionaries are leaders.

SECTION 3 — VARC (3 Questions)

Q16. Grammar / Sentence Correction
Identify the error in the sentence:
"Neither the manager nor the employees was present at the meeting."

Q17. Para-Jumble
Arrange sentences P, Q, R, S into a coherent paragraph:

- **P:** However, the most profound limitation of this technology lies not in its current capability but in what it reveals about human nature.
- **Q:** Artificial intelligence has made remarkable strides in mimicking human creativity.
- **R:** We seem to want machines to be creative so that we can better understand what creativity means.
- **S:** Systems can now compose music, generate art, and even write poetry that fools trained eyes.

Q18. Reading Comprehension — Inference
"The paradox of choice suggests that having more options leads to worse decisions, not better ones. When presented with too many alternatives, individuals experience decision fatigue, often leading to regret regardless of their final choice. Psychologist Barry Schwartz argues that the maximizer — someone who seeks the best possible option — suffers more than the satisficer — someone who settles for 'good enough.'"

Which of the following would the author most likely agree with?
(A) More product variety always improves customer satisfaction.
(B) People who set high standards tend to be more content.
(C) Limiting options can sometimes lead to greater happiness.
(D) Decision fatigue is an irreversible condition.

How to Respond
Post your answers in this format:

Q1: [your answer] | Working: [your steps]
Q2: [your answer] | Working: [your steps]
... and so on

Mixtures & Alligations

Time, Speed & Distance

Trains & Boats

Time & Work

Pipes & Cisterns

Averages

MODULE 3 — ALGEBRA

Module 3

Algebra

~20% of QA · 10-12 questions

Linear equations

Quadratic equations

Inequalities

Logarithms

Surds & Indices

AP, GP, HP

Functions & Graphs

Polynomials

MODULE 4 — GEOMETRY & MENSURATION

Module 4

Geometry & Mensuration

~20% of QA · 10-12 questions

Lines & angles

Triangles

Circles

Quadrilaterals

Polygons

Coordinate geometry

2D mensuration

3D mensuration

Basic trigonometry

MODULE 5 — MODERN MATH

Module 5

Modern Math

~15% of QA · 6-8 questions

Permutations

Combinations

Probability

Set Theory

Venn diagrams

Starting now → Module 1, Topic 1: Types of Numbers

Each concept is taught to full depth before moving on. No concept skipped.

MODULE 1 — NUMBER SYSTEMS

Chapter 1: The Number Universe

Everything in QA sits on top of this. If your number sense is sharp, half of CAT becomes instinct.

Module 2

Arithmetic

~40% of QA · 20-25 questions

Percentages

Profit & Loss

Simple Interest

Compound Interest

Ratio & Proportion

Mixtures & Alligations

Time, Speed & Distance

Trains & Boats

Time & Work

Pipes & Cisterns

Averages

MODULE 3 — ALGEBRA

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Algebra

~20% of QA · 10-12 questions

Linear equations

Quadratic equations

Inequalities

Logarithms

Surds & Indices

AP, GP, HP

Functions & Graphs

Polynomials

MODULE 4 — GEOMETRY & MENSURATION

Module 4

Geometry & Mensuration

~20% of QA · 10-12 questions

Lines & angles

Triangles

Circles

Quadrilaterals

Polygons

Coordinate geometry

2D mensuration

3D mensuration

Basic trigonometry

MODULE 5 — MODERN MATH

Module 5

Modern Math

~15% of QA · 6-8 questions

Permutations

Combinations

Probability

Set Theory

Venn diagrams

Starting now → Module 1, Topic 1: Types of Numbers

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MODULE 1 — NUMBER SYSTEMS

Chapter 1: The Number Universe

Everything in QA sits on top of this. If your number sense is sharp, half of CAT becomes instinct.

All rationals + all Irrationals combined. Everything on the number line.

Integers specifically for CAT — Even and Odd:

Even numbers: divisible by 2. {..., -4, -2, 0, 2, 4, ...}

Odd numbers: not divisible by 2. {..., -3, -1, 1, 3, 5, ...}

Rules that save time in CAT:

Operation	Result
Even + Even	Even
Odd + Odd	Even
Even + Odd	Odd
Even × Even	Even
Even × Odd	Even
Odd × Odd	Odd

1.2 — Prime Numbers (Extremely High CAT Weightage)

A prime number has exactly two factors: 1 and itself.

Primes up to 100 — memorize these cold:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

There are exactly 25 primes below 100.

- Critical facts:**
- 1 is NOT prime (it has only one factor: itself)
 - 2 is the ONLY even prime
 - All primes > 3 are of the form $6k \pm 1$ (but not all $6k \pm 1$ are prime — e.g. $25 = 6 \times 4 + 1$, not prime)

How to check if a number N is prime:

Test divisibility by all primes up to $\sqrt{N}$. If none divide N, it's prime.

Example: Is 127 prime?

$\sqrt{127} \approx 11.3$, so test primes up to 11: 2, 3, 5, 7, 11.

$127 \div 2 \rightarrow$ not integer. $127 \div 3 \rightarrow 42.3$, no. $127 \div 5 \rightarrow 25.4$, no. $127 \div 7 \rightarrow 18.1$, no. $127 \div 11 \rightarrow 11.5$, no.

127 is prime.

1.3 — Divisibility Rules (Speed = Marks)

These eliminate the need to divide. Know them without thinking.

Divisible by 2: Last digit is 0, 2, 4, 6, or 8.

Example: 4738 $\rightarrow$ last digit 8 $\rightarrow$ yes.

Divisible by 3: Sum of all digits is divisible by 3.

Example: 4738 $\rightarrow 4+7+3+8 = 22 \rightarrow 22 \div 3 = 7.3 \rightarrow$ No.

Example: 4731 $\rightarrow 4+7+3+1 = 15 \rightarrow 15 \div 3 = 5 \rightarrow$ Yes.

Divisible by 4: Last two digits form a number divisible by 4.

Example: 3748 $\rightarrow$ last two digits = 48 $\rightarrow 48 \div 4 = 12 \rightarrow$ Yes.

Example: 3746 $\rightarrow 46 \div 4 = 11.5 \rightarrow$ No.

Divisible by 5: Last digit is 0 or 5.

Divisible by 6: Divisible by both 2 AND 3.

Divisible by 7: No clean rule — just divide. (There is a rule but it's slower than dividing for 2-digit numbers.)

Divisible by 8: Last three digits form a number divisible by 8.

Example: 54872 $\rightarrow 872 \div 8 = 109 \rightarrow$ Yes.

Divisible by 9: Sum of all digits is divisible by 9.

Example: 54872 $\rightarrow 5+4+8+7+2 = 26 \rightarrow 26 \div 9 = 2.8 \rightarrow$ No.

Example: 54819 $\rightarrow 5+4+8+1+9 = 27 \rightarrow 27 \div 9 = 3 \rightarrow$ Yes.

Divisible by 10: Last digit is 0.

Divisible by 11: (Sum of digits at odd positions) – (Sum of digits at even positions) = 0 or multiple of 11.

Example: 83215

Odd positions (1st, 3rd, 5th from left): 8, 2, 5 $\rightarrow$ sum = 15

Even positions (2nd, 4th): 3, 1 $\rightarrow$ sum = 4

Difference: $15 - 4 = 11 \rightarrow$ divisible by 11. Yes.

Divisible by 12: Divisible by both 3 AND 4.

Divisible by 25: Last two digits divisible by 25 (i.e., last two digits are 00, 25, 50, or 75).

1.4 — HCF and LCM (High frequency in CAT)

HCF (Highest Common Factor) — also called GCD.

The largest number that divides both (all) given numbers.

LCM (Lowest Common Multiple)

The smallest number that is divisible by both (all) given numbers.

The most important relationship:

For any two numbers A and B:

$$| \quad \text{HCF} \times \text{LCM} = A \times B$$

This holds for exactly two numbers. Memorize it — it's used in 70% of HCF/LCM problems.

How to find HCF — Prime Factorization method:

HCF = product of common prime factors with their lowest powers.

Example: HCF(72, 120)

$$72 = 2^3 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

Common factors: 2^3 and 3^1

$$\text{HCF} = 2^3 \times 3 = 8 \times 3 = \mathbf{24}$$

How to find LCM — Prime Factorization method:

LCM = product of all prime factors with their highest powers.

Example: LCM(72, 120)

$$72 = 2^3 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 8 \times 9 \times 5 = \mathbf{360}$$

$$\text{Verify: } \text{HCF} \times \text{LCM} = 24 \times 360 = 8640 = 72 \times 120 \quad \checkmark$$

CAT Application Types:

Type 1 — "Find the largest number that divides X and Y leaving the same remainder r."

Answer: $\text{HCF}(X-r, Y-r)$

Type 2 — "Find the largest number that divides X, Y, Z leaving remainders a, b, c."

Answer: $\text{HCF}(X-a, Y-b, Z-c)$

Type 3 — "Find the smallest number divisible by X, Y, Z."

Answer: $\text{LCM}(X, Y, Z)$

Type 4 — "Find the smallest number that when divided by X, Y, Z leaves remainder r each time."

Answer: $\text{LCM}(X, Y, Z) + r$

1.5 — Factors and Number of Factors

Prime Factorization — the core tool.

Every composite number can be written uniquely as a product of primes.

$$\text{Example: } 360 = 2^3 \times 3^2 \times 5^1$$

Formula for Number of Factors:

$$\text{If } N = p^a \times q^b \times r^c \times \dots$$

$$\text{Then number of factors} = (a+1)(b+1)(c+1)\dots$$

Example: How many factors does 360 have?

$$360 = 2^3 \times 3^2 \times 5^1$$

$$\text{Number of factors} = (3+1)(2+1)(1+1) = 4 \times 3 \times 2 = \mathbf{24}$$

Sum of all factors:

Example: LCM(72, 120)

$$72 = 2^3 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 8 \times 9 \times 5 = \mathbf{360}$$

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Example: How many factors does 360 have?

$360 = 2^3 \times 3^2 \times 5^1$

Number of factors = $(3+1)(2+1)(1+1) = 4 \times 3 \times 2 = \mathbf{24}$

Sum of all factors:	$a, b, c, d, \dots$	1
7	7, 9, 3, 1, 7, 9, 3, 1...	4
8	8, 4, 2, 6, 8, 4, 2, 6...	4
9	9, 1, 9, 1...	2
0	0, 0, 0, 0...	1

How to use:

Unit digit of 7^{53} ?

Cycle of 7: [7, 9, 3, 1] — period 4.

$53 \div 4 = 13$ remainder 1.

Remainder 1 → 1st position in cycle → 7.

Unit digit of 2^{100} ?

$100 \div 4 = 25$, remainder 0.

Remainder 0 → take last position in cycle → 6.

1.7 — Remainders (The deep topic)

Basic remainder: When N is divided by D, remainder = $N - D \times \lfloor N/D \rfloor$

Key property: $(a + b) \bmod D = ((a \bmod D) + (b \bmod D)) \bmod D$

This means you can find remainders of each part separately, then add.

Example: Find remainder when 2^{25} is divided by 7.

Powers of 2 mod 7: $2^1=2, 2^2=4, 2^3=8 \rightarrow 1, 2^4=2, 2^5=4, 2^6=1 \dots$

The cycle is [2, 4, 1] — period 3.

$25 \div 3 = 8$ remainder 1.

Remainder 1 → 1st position → 2.

Fermat's Little Theorem (CAT level):

If p is prime and $\gcd(a, p) = 1$, then:

$a, b, c, d, \dots$	1	
7	7, 9, 3, 1, 7, 9, 3, 1...	4
8	8, 4, 2, 6, 8, 4, 2, 6...	4
9	9, 1, 9, 1...	2
0	0, 0, 0, 0...	1

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Remainder 1 → 1st position → 2.

Fermat's Little Theorem (CAT level):

If p is prime and $\gcd(a, p) = 1$, then:

Q3 (HCF/LCM): The LCM of two numbers is 180 and their HCF is 12. One number is 36. Find the other.

Q4 (Factors): How many factors does 7200 have?

Q5 (Unit digit): Find the unit digit of $3^{473} + 7^{54}$.

Q6 (Remainder): Find the remainder when 5^{99} is divided by 6.

Q7 (CAT level): Find the number of trailing zeros in 200!

Q8 (CAT level): What is the largest power of 2 that divides 50!?

Q9 (CAT level): A number when divided by 6 gives remainder 4. When the same number is divided by 18, what is the possible remainder?

Show me your working for each. Write everything — even the thought process. That's where

I find the gaps.

Once you attempt these, I evaluate each answer, explain every mistake, identify your exact weak points, and then we move to Arithmetic (which is where 40% of CAT QA lives).

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The PDF covers all 9 sections of Module 1 with a clean layout — teal headers, colour-coded info boxes, trap warnings, speed tips, and formula blocks. Each section has worked examples, and the last two pages contain all 9 practice questions with working space, plus a one-page quick reference summary of every formula from the module.

When you're done attempting the practice questions, share your working here and I'll evaluate each answer, identify your exact weak spots, and we move to Module 2 — Arithmetic.

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 6. Mixtures & Alligations

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Module 2 Arithmetic PDF generator

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